



Incident report analysis

Summary	The organization experienced a denial-of-service (DoS) attack that caused internal network services to become unavailable for approximately two hours. A malicious actor flooded the company's network with ICMP packets through an unconfigured firewall, overwhelming network resources and preventing normal internal traffic from accessing network services. The incident management team responded by blocking incoming ICMP packets, taking non-critical network services offline, and restoring critical services. Following the incident, the network security team implemented ICMP rate limiting, source IP address verification, network monitoring, and IDS/IPS controls to reduce the risk and impact of similar attacks in the future.
Identify	The incident was a denial-of-service (DoS) attack caused by a flood of incoming ICMP packets. The attacker was able to inject the traffic into the organization's internal network because of an unconfigured firewall. The attack flooded the network resources and blocked normal internal traffic from reaching network services for about two hours. The primary security vulnerability found was the absence of suitable firewall rules in place to handle inbound ICMP traffic.
Protect	The organization should properly configure and regularly audit its firewall to block or rate-limit unauthorized ICMP traffic. Source IP verification should be enabled to help prevent spoofed traffic. The IDS/IPS and network monitoring systems should also be maintained and updated to provide additional protection against suspicious traffic. Regular security audits of firewall rules and network devices should be performed to identify and fix configuration weaknesses.

Detect	The organization should continuously monitor network traffic using network monitoring software, firewall logs, and an IDS/IPS. Alerts should be configured to detect unusual increases in ICMP traffic, traffic from untrusted or suspicious IP addresses, and other abnormal network patterns. A SIEM can also be used to collect and analyze security logs, helping the security team quickly identify and investigate potential incidents.
Respond	The organization should follow an incident response plan to quickly contain the attack by blocking or rate-limiting malicious ICMP traffic and isolating affected network services when necessary. The security team should analyze firewall, IDS/IPS, and network logs to determine the source and characteristics of the attack. The incident should be documented and communicated to relevant IT staff and management, and the response procedures should be reviewed for future improvements.
Recover	The organization should restore critical network services and gradually return non-critical services to normal operation after confirming that the attack has been contained. Network connectivity and firewall configurations should be verified to ensure systems are functioning securely. The team should document the incident, review recovery procedures, and update security controls based on lessons learned to improve future recovery efforts.

Reflections/Notes: This incident helped me understand how the NIST Cybersecurity Framework can be applied to a real-world network security incident. I learned that properly configured firewalls, continuous monitoring, and intrusion detection can help prevent and detect DoS attacks. I also learned that having clear response and recovery procedures is important for minimizing downtime and restoring critical services quickly.